# AMPLENESS DESCENDS ALONG A QUASI-COMPACT MORPHISM: FROM THE COMPOSITE TO THE FIRST FACTOR

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*Allowed standard prerequisites (editorial).*

- Ampleness restricts to a locally closed subscheme. Properties, Lemma \ref{properties-lemma-ample-on-locally-closed}, cited at proof lines 8947-8948. Transfers ampleness to the smaller affine open.
- The preimage of an affine open is a quasi-compact open immersion when the morphism is quasi-compact over a separated target. Schemes, Lemma \ref{schemes-lemma-quasi-compact-permanence}, cited at proof lines 8949-8953, together with Properties, Lemma \ref{properties-lemma-ample-separated}. Supplies the quasi-compactness and separation used to apply the previous lemma.
- The characterisation of relative ampleness by an affine open covering of the base. Lemma \ref{lemma-characterize-relatively-ample}, cited at proof lines 8956-8958. Turns ampleness on each preimage into relative ampleness.

*Source reference.* Original lemma 0C4L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15231–15231. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

*Source reference.* Original lemma 03GI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4529–4529. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 0B3E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13962–13962. Navigation pointer to a statement the

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*Source reference.* Original lemma 09MP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12187–12187. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 01VJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2448–2448. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

**Lemma 0.2.** *Let $g : Y \to S$ and $f : X \to Y$ be morphisms of schemes. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. If $\mathcal{L}$ is $g \circ f$-ample and $f$ is quasi-compact*[1] *then $\mathcal{L}$ is $f$-ample.*

*Proof.* Assume $f$ is quasi-compact and $\mathcal{L}$ is $g \circ f$-ample. Let $U \subset S$ be an affine open and let $V \subset Y$ be an affine open with $g(V) \subset U$. Then $\mathcal{L}|_{(g\circ f)^{-1}(U)}$ is ample on $(g \circ f)^{-1}(U)$ by assumption. Since $f^{-1}(V) \subset (g \circ f)^{-1}(U)$ we see that $\mathcal{L}|_{f^{-1}(V)}$ is ample on $f^{-1}(V)$ by Properties, Lemma 0B3E. Namely, $f^{-1}(V) \to (g \circ f)^{-1}(U)$ is a quasi-compact open immersion by Schemes, Lemma 03GI as $(g \circ f)^{-1}(U)$ is separated (Properties, Lemma 09MP) and $f^{-1}(V)$ is quasi-compact (as $f$ is quasi-compact). Thus we conclude that $\mathcal{L}$ is $f$-ample by Lemma 01VJ. □

[1] This follows if $g$ is quasi-separated by Schemes, Lemma 03GI.